

Total No. of Questions : 8]

SEAT No. :

PC2503

[Total No. of Pages : 2

[6354]-633

B.E. (Mechanical)

PROCESS EQUIPMENT DESIGN

(2019 Pattern) (Semester - VIII) (Elective - VI) (402051A)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data wherever necessary.

- Q1)** a) What are the types of vessels and explain classes of vessels. [9]
b) Explain vessel opening compensation method. [8]

OR

- Q2)** a) Determine equivalent stress under combined loading by considering seismic loads and wind loads. [10]
b) Explain with sketch Pre stressing of thick cylinder. [7]

- Q3)** a) Explain with neat sketch design of storage vessels. [10]
b) Explain with neat sketch design of reaction vessels. [8]

OR

- Q4)** a) Explain the selection of pumps and compressors. [10]
b) Define and explain types of valves used on pipe line. [8]

- Q5)** a) Define processes, process parameters and their correlation. [10]
b) Explain fundamentals of process measurements. [7]

OR

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- Q6)** a) Explain with sketch control modern control devices. [9]
b) Explain controls of major unit operation and processes. [8]
- Q7)** a) Explain inspection and erection of process equipment like pressure vessels and chimneys. [10]
b) Explain inspection and erection of process equipment like ducting and heat exchangers. [8]

OR

- Q8)** a) Explain planning and manufacture of process equipment like pressure vessels and chimneys. [10]
b) Explain fuel pumping stations. [8]

